OBJECT™
TECHNOLOGIES

Process Management

Process management

- Defination of Process
- Process Life Cycle
- Process Synchronization
- Introduction to Linus
- Basic commands of Linux

Defination of process

- Process is....... "Program in execution."
- It is a part of program which is under execution.
- Program is a passive entity.
- Process is a active entity.

Note :: Any program or process which is executed by the system is to be brought into the primary memory(RAM).

Memory structure used by C/Java code

Stack :: The Stack contains the temporary data such as method/function, parameters, return address and local variables.

Data :: This section contains the global and static variables.

Text :: This includes the current activity represented by the value of Program Counter and the contents of the processor's registers.

Heap :: This is dynamically allocated memory to a process during its run time. This memory can be released once execution is completed.

Memory Allocation Process

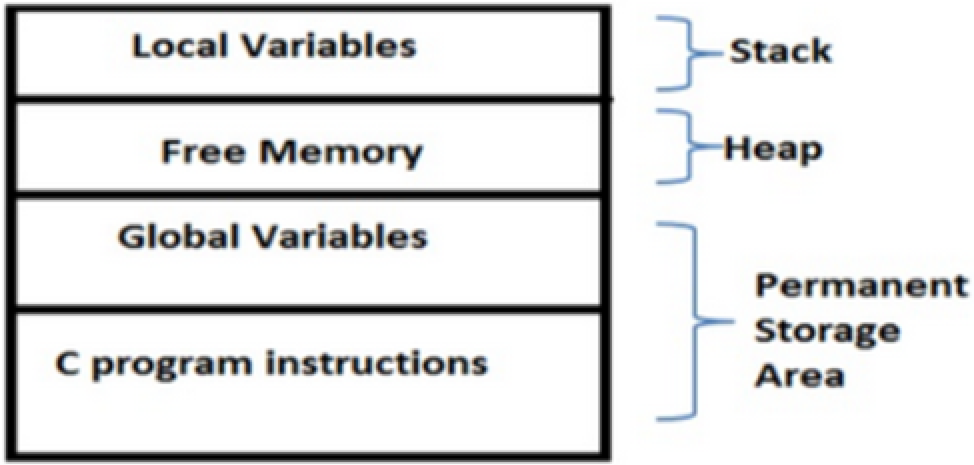
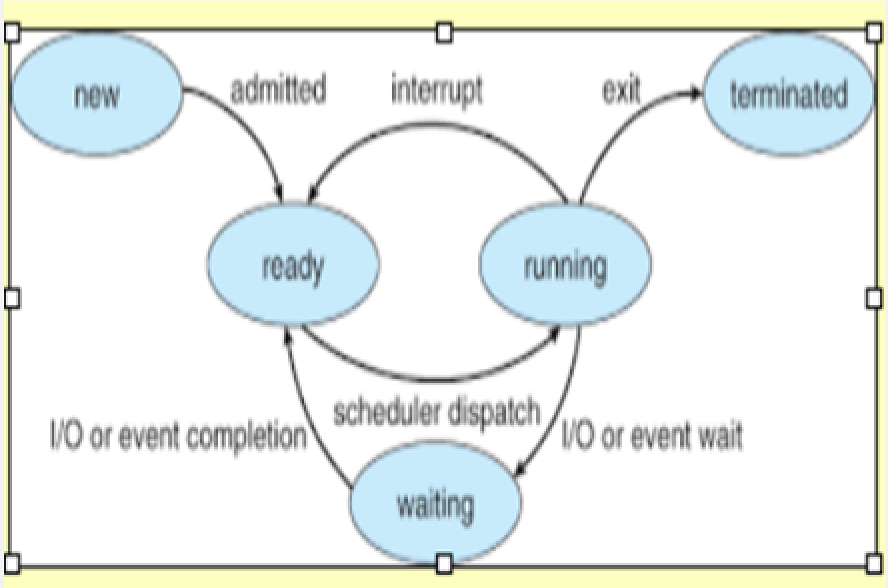


Fig: Storage of a C program

States of Process

Any Process falls under following states::

1. **New** :: Process is in New State means a new process is created , and it is yet not active.
2. **Ready** :: Process comes in RAM and gets placed in a Queue. This queue is called as Ready Queue. This task is done by long term scheduler.
3. **Running** :: Process in RAM gets CPU for execution. This task of dispatching process from Ready Queue to CPU is done by short term scheduler also called as Dispatcher.
4. **Terminate** :: Process is completed logically. Hence process is terminated. Now system has no record of this process.
5. **Wait** :: Sometimes when process is in execution phase, it may require some other resources to complete the execution (mostly these are i/o resources. Speed of i/o resources is very slow hence CPU becomes ideal). System puts this process in the wait state and CPU starts executing another process. Once i/o operation of process is completed, process again goes in Ready Queue and waits for CPU.



Process Control Block

-- PCB stores detailed information about Process .

-- Every process has it's separate PCB.

-- The information of the Process is used by CPU at the Run time .

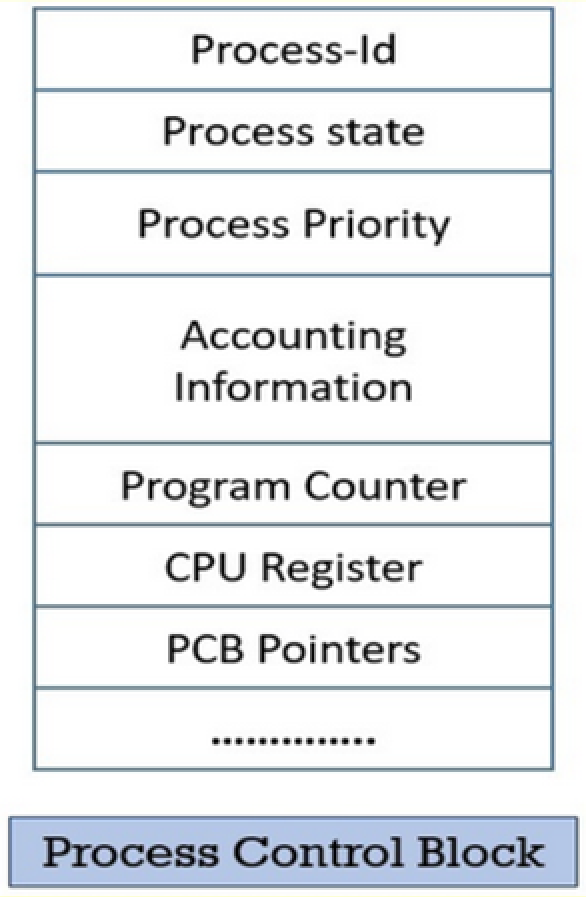
-- PCB is also called as Task Control Block.

-- The PCB is identified by an unique integer called as Process ID (PID).

-- System also maintains a table of all these PCBs , to keep track of all running processes.

-- Once process goes in terminate state, it's entry from PCB is deleted.

--Helpful in context switching.



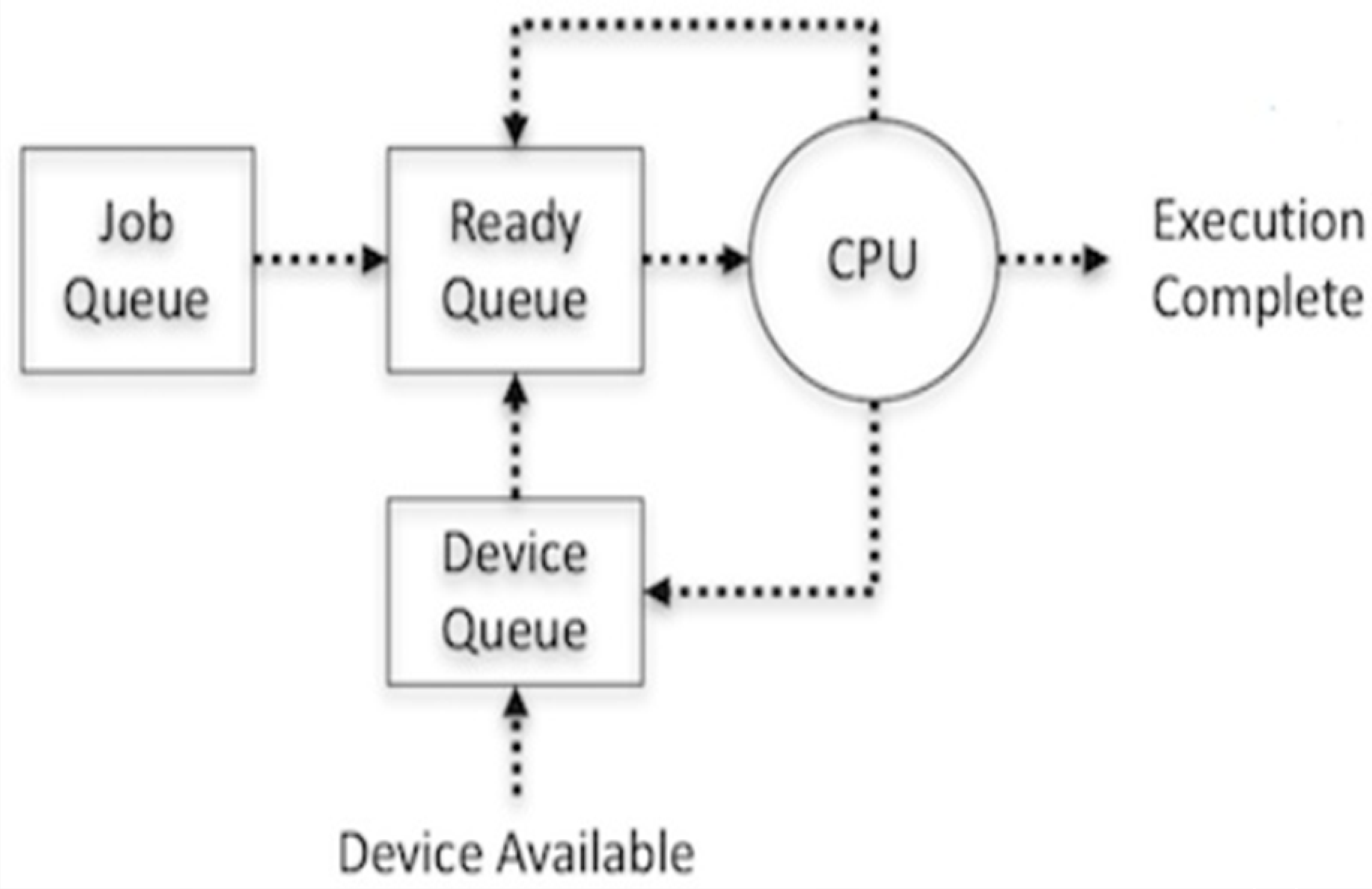
Types of Processes

- I/O Bound Process

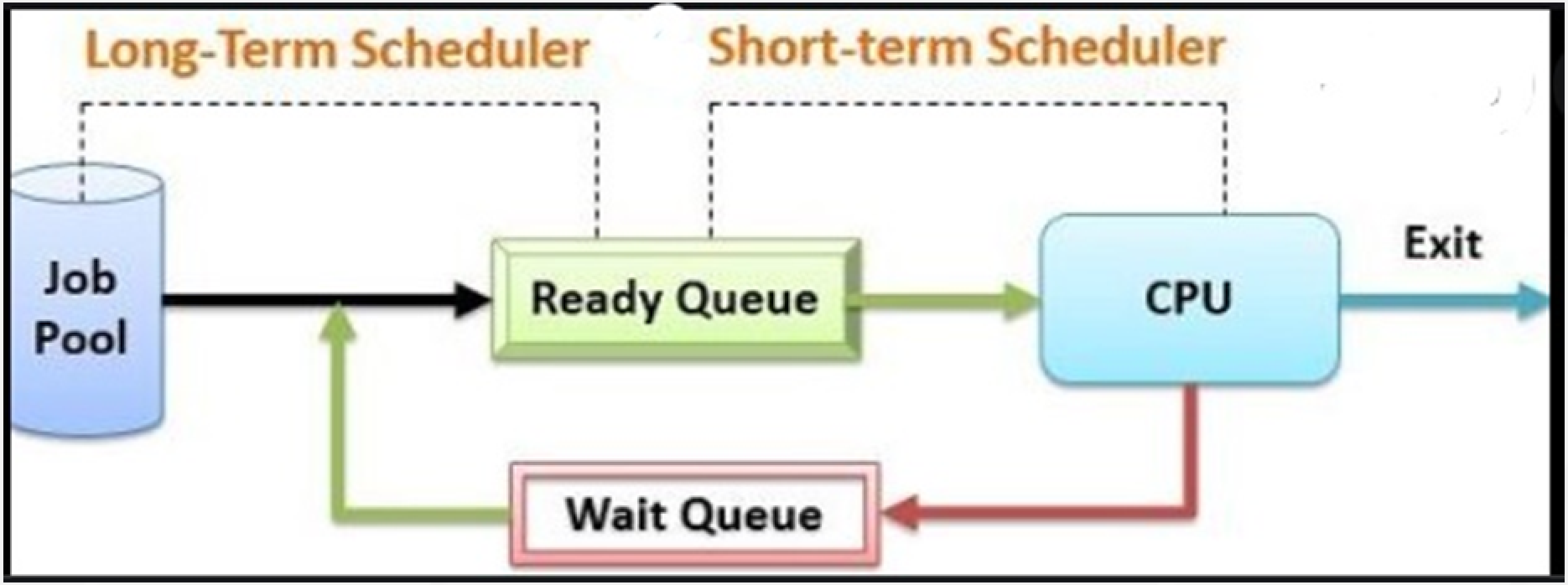
  -Spends more time doing I/O than computations.

- CPU Bound Process

  -Spends more time doing computations than I/O.


Job
Queue
Ready
Queue
CPU
Execution
Complete
Device
Queue
Device Available

Schedulers in OS

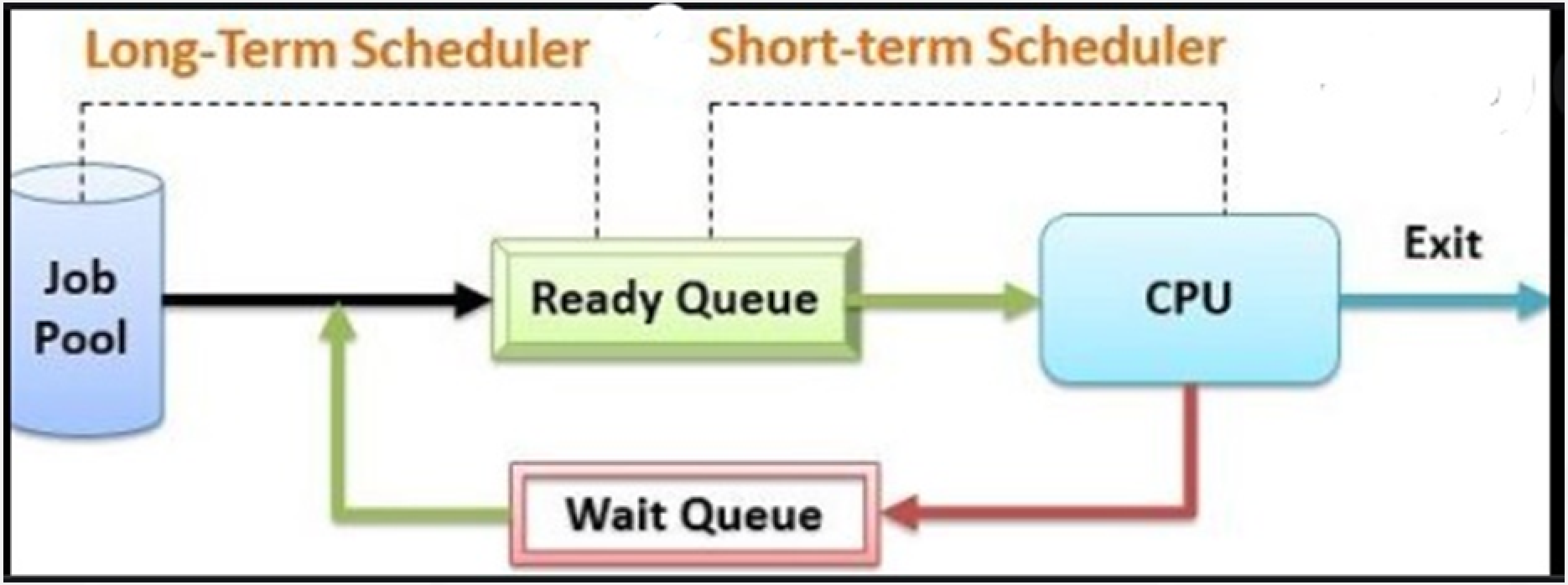


1. Long Term Scheduler::

-When a new process is created it enters Job Queue.

- Depending on the load in Ready Queue , Long Term Scheduler takes the process from Job Queue to Ready Queue.

Schedulers in OS

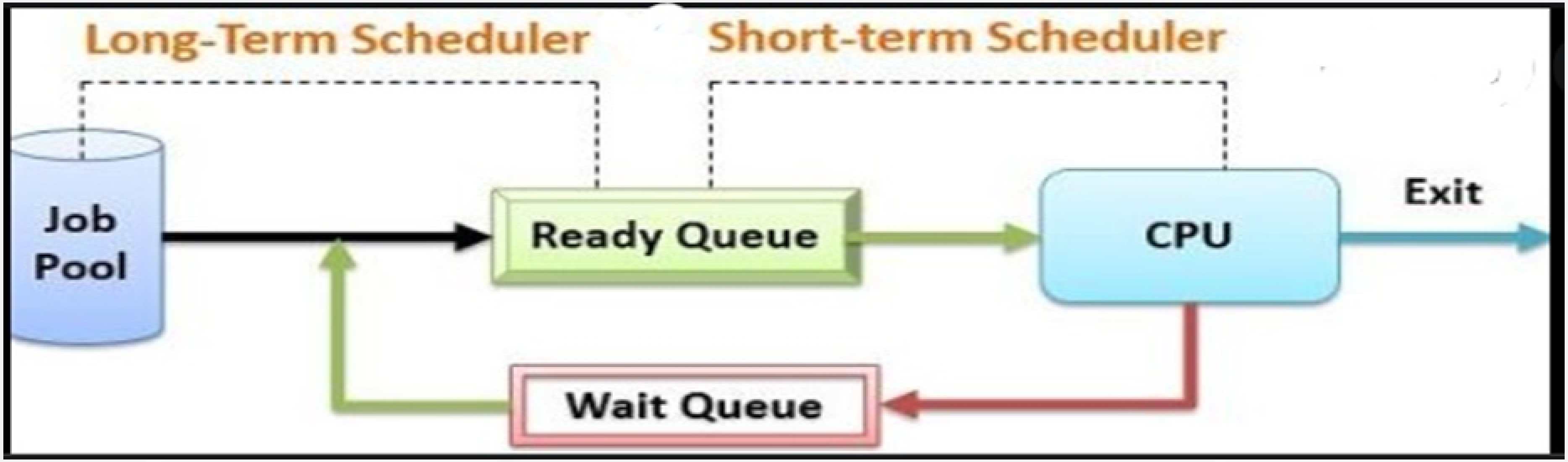


2. Short Term Scheduler ::

-This code of OS picks the process from ready queue and assigns it to CPU for execution.

- Short term scheduler is also called as Dispatcher. .

Schedulers in OS



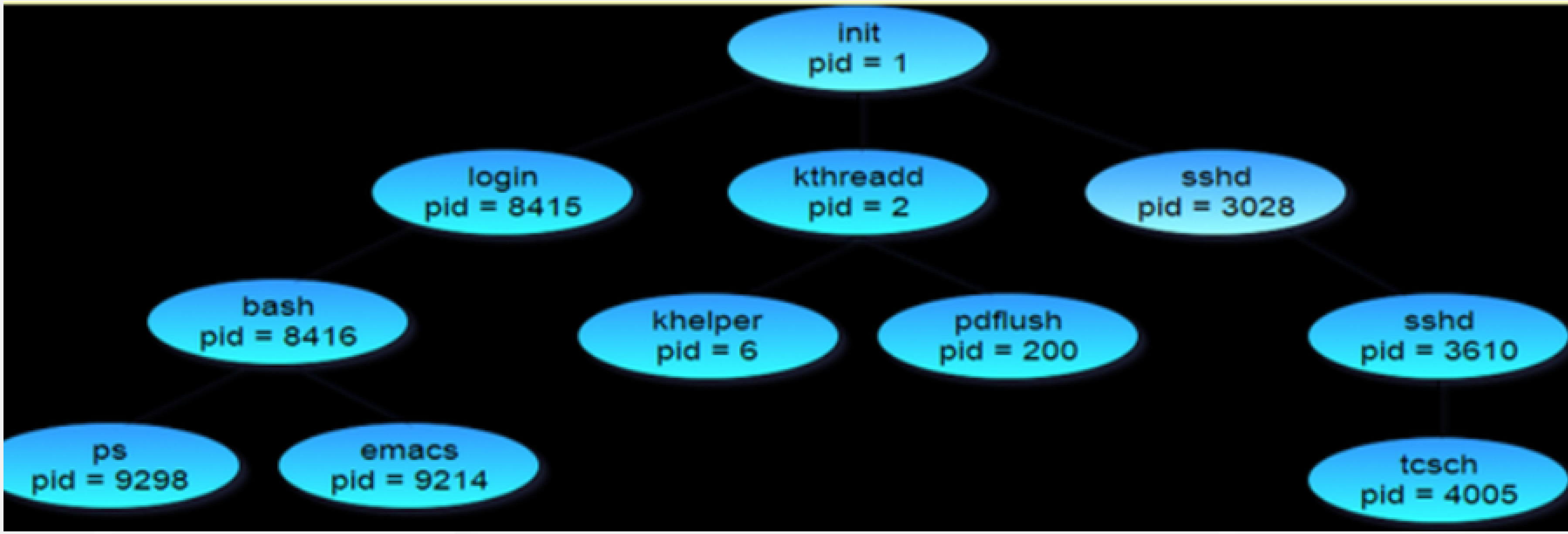
3. Medium Term Scheduler ::

- Sometimes Ready Queue gets overloaded by processes .
- Med Term scheduler creates a **Swap Area** on the hard disk of the system and moves some processes on the disk .
- This technique of moving processes from RAM to hard disk is called as **Swap-Out**.
- After Ready Queue gets lighter load ,the processes are moved back again ot RAM.
- This is called as **Swap-In** technique.
- Medium Term Scheduler will come into picture only when Ready Queue gets loaded with too many processes.

- Degree of **Multiprogramming** ::

  - Number of processes in the Ready Queue actively waiting for CPU defines the degree of multiprogramming.

  - More the degree of multiprogramming , more is the load on short term scheduler but CPU utilization is optimum in that case.



--Process creation is system call to the kernel.

--**fork()** command in linux is used to create a new process.

--When the system starts(boots) ,'init' is the first process created by OS. It has pid=1 . This process is father of all the processes created there after.

--This process is the last to terminate.

Process Termination::

Whenever any process terminats, it follows certain methodology --

-It returns delivery status to parent process.

-It releases all the resources which it was holding .

-It deletes its PCB from Process Control Table.

1. Cascading Effect::

-Sometimes parent processes is killed irrespective of running child processes.

-In such case child processes are forcefully terminated.

-So all the processes running under parent are terminated.

-This is called as cascading effect.

2. Orphan Process ::

-Sometimes if parent process is terminated, and child processes still remains, it is called as Orphan Process.

-In this case, the orphan processes are assigned to main 'init' process.

3. Zombie Process ::

-Sometimes though the process terminates , it's PCB remains in the memory.

-This is called as Zombi process.